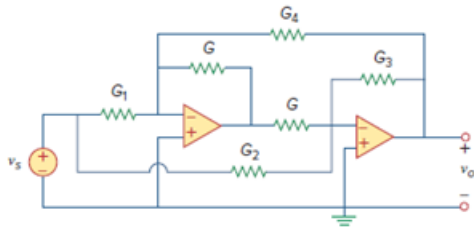


Problem

For the op amp circuit shown in Fig. 5.91, find v_o/v_s .

Figure. 5.91:



Step-by-step solution

Step 1 of 6

Refer to Figure 5.91 in the text book.

In an ideal op-amp, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal.

The modified circuit diagram is shown in Figure 1.

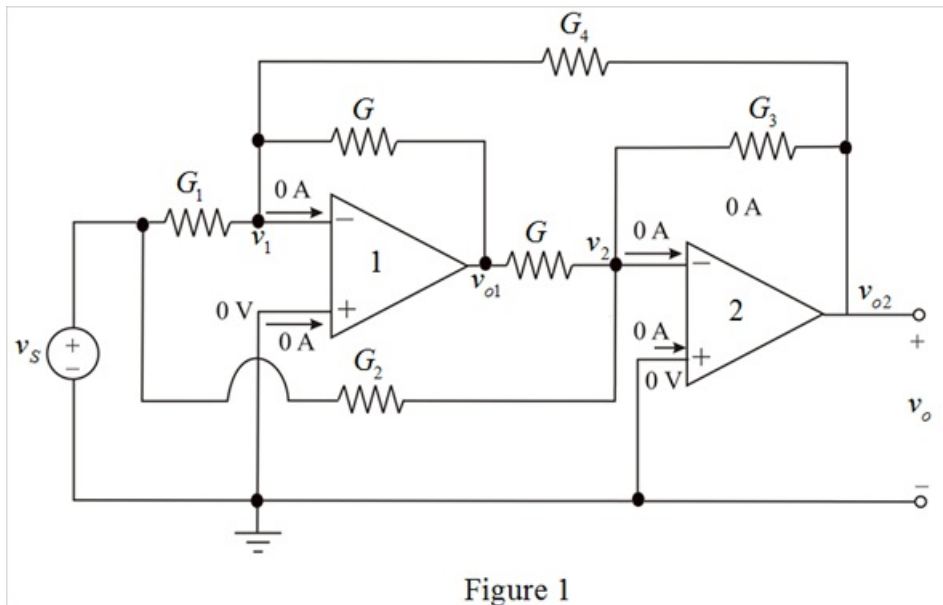


Figure 1

Step 2 of 6

From Figure 1, the inverting terminal of the first operational amplifier is,

$$\begin{aligned} v_1 &= v^- \\ &= v^+ \\ &= 0 \text{ V} \end{aligned}$$

From Figure 1, the inverting terminal of the second operational amplifier is,

$$\begin{aligned} v_2 &= v^- \\ &= v^+ \\ &= 0 \text{ V} \end{aligned}$$

Step 3 of 6

Apply Kirchhoff's Current law at node v_1 in Figure 1.

$$\frac{v_1 - v_s}{\left(\frac{1}{G_1}\right)} + \frac{v_1 - v_{o1}}{\left(\frac{1}{G}\right)} + \frac{v_1 - v_{o2}}{\left(\frac{1}{G_4}\right)} = 0$$

Substitute 0 V for v_1 in the equation.

$$\frac{0 - v_s}{\left(\frac{1}{G_1}\right)} + \frac{0 - v_{o1}}{\left(\frac{1}{G}\right)} + \frac{0 - v_{o2}}{\left(\frac{1}{G_4}\right)} = 0$$

$$(0 - v_s)G_1 + (0 - v_{o1})G + (0 - v_{o2})G_4 = 0$$

$$-G_1v_s - Gv_{o1} - G_4v_{o2} = 0$$

$$Gv_{o1} = -G_1v_s - G_4v_{o2}$$

$$v_{o1} = \left(\frac{-G_1v_s - G_4v_{o2}}{G} \right)$$

$$v_{o1} = -\left(\frac{G_1}{G}\right)v_s - \left(\frac{G_4}{G}\right)v_{o2}$$

Step 4 of 6

Apply Kirchhoff's Current law at node v_2 in Figure 1.

$$\frac{v_2 - v_{o1}}{\left(\frac{1}{G}\right)} + \frac{v_2 - v_s}{\left(\frac{1}{G_2}\right)} + \frac{v_2 - v_{o2}}{\left(\frac{1}{G_3}\right)} = 0$$

Substitute 0 V for v_2 in the equation.

$$\frac{0 - v_{o1}}{\left(\frac{1}{G}\right)} + \frac{0 - v_s}{\left(\frac{1}{G_2}\right)} + \frac{0 - v_{o2}}{\left(\frac{1}{G_3}\right)} = 0$$

$$(0 - v_{o1})G + (0 - v_s)G_2 + (0 - v_{o2})G_3 = 0$$

$$-Gv_{o1} - G_2v_s - G_3v_{o2} = 0$$

$$G_3v_{o2} = -Gv_{o1} - G_2v_s$$

$$v_{o2} = -\left(\frac{G}{G_3}\right)v_{o1} - \left(\frac{G_2}{G_3}\right)v_s \dots\dots (1)$$

Step 5 of 6

Recall equation (1).

$$v_{o2} = -\left(\frac{G}{G_3}\right)v_{o1} - \left(\frac{G_2}{G_3}\right)v_s$$

Substitute $-\left(\frac{G_1}{G}\right)v_s - \left(\frac{G_4}{G}\right)v_{o2}$ for v_{o1} in equation (1) to determine the voltage gain.

$$v_{o2} = -\left(\frac{G}{G_3}\right)\left[-\left(\frac{G_1}{G}\right)v_s - \left(\frac{G_4}{G}\right)v_{o2}\right] - \left(\frac{G_2}{G_3}\right)v_s$$

$$v_{o2} = \left(\frac{G}{G_3}\right)\left(\frac{G_1}{G}\right)v_s + \left(\frac{G}{G_3}\right)\left(\frac{G_4}{G}\right)v_{o2} - \left(\frac{G_2}{G_3}\right)v_s$$

$$v_{o2} = \left(\frac{G_1}{G_3}\right)v_s + \left(\frac{G_4}{G_3}\right)v_{o2} - \left(\frac{G_2}{G_3}\right)v_s$$

$$v_{o2} - \left(\frac{G_4}{G_3}\right)v_{o2} = \left(\frac{G_1}{G_3}\right)v_s - \left(\frac{G_2}{G_3}\right)v_s$$

Further simplification as follows:

$$v_{o2} \left[1 - \frac{G_4}{G_3} \right] = \left(\frac{G_1}{G_3} - \frac{G_2}{G_3} \right) v_s$$

$$v_{o2} \left[\frac{G_3 - G_4}{G_3} \right] = \left(\frac{G_1 - G_2}{G_3} \right) v_s$$

$$v_{o2} [G_3 - G_4] = (G_1 - G_2) v_s$$

$$v_{o2} = \left(\frac{G_1 - G_2}{G_3 - G_4} \right) v_s$$

$$\frac{v_{o2}}{v_s} = \left(\frac{G_1 - G_2}{G_3 - G_4} \right)$$

$$\frac{v_o}{v_s} = \left(\frac{G_1 - G_2}{G_3 - G_4} \right) \quad (\text{since } v_{o2} = v_o)$$

Thus, the voltage gain of the op-amp circuit as shown in Figure 5.91 is,

$$\boxed{\frac{v_o}{v_s} = \left(\frac{G_1 - G_2}{G_3 - G_4} \right)}$$